Business Model Canvas: Energy Supply Systems (Module 05)

Infineon/BIT Excellence Program 2026 | Sustainable dried mango supply chain, Burkina Faso to Germany | Prepared by MADIEGA S Aida Justine

Key Partners

- Other modules: Water Supply (pump), Product Transformation (dryer), Machinery
- SONABEL (grid emission factor)
- Solar and battery equipment suppliers (to validate)

Key Activities

- Monitor solar production
- Log consumption and generator events
- Calculate CO_2 avoided
- Simulate expansion scenarios (What-If)
- Raise alerts

Key Resources

- PV array: 8.24 kWp
- Battery bank: 42.5 kWh
- Backup diesel generator: 8 kW
- Inverter and IoT monitoring controller
- Database with sourced reference data

Value Proposition

- Reliable energy for the whole farm: solar first, battery and diesel as backup
- Less diesel and lower energy cost
- Measurable carbon footprint (about 539 kg CO_2 avoided per month, base case)
- What-If tool to size the next expansion

Customer Relationships

- Self-service dashboard
- Automatic alerts
- Data exchange with other modules through internal events (planned)

Channels

- Energy dashboard (7 tabs)
- REST API used by other modules
- Alerts and notifications

Customer Segments

- Internal modules that consume energy
- Farm operator
- Downstream: German buyers who value a low-carbon supply chain (to validate)

Cost Structure

- PV: about EUR 9,891 (8.24 kWp x EUR 1,200/kWp)
- Battery: about EUR 17,000 (42.5 kWh x EUR 400/kWh)
- Generator fuel and maintenance , PV upkeep
- Note: the battery price is above current West African market prices; kept by team convention (EUR)

Revenue Streams / Value Captured

- No direct sales: this module creates value through avoided costs (diesel, grid)
- Carbon data that supports the sustainability claim for export (to validate)
- Open point, is energy a shared internal cost or a service billed to other modules?

Supporting notes

Sourced figures

- Base consumption: 34 kWh/day.
- PV capacity: 8.24 kWp = 34 / (5.5 peak sun hours x 0.75 performance ratio). Source : Nana et al. 2023, Clean Air Journal 33(2), citing MEEVCC 2018.
- CO_2 avoided: 538.80 kg/month, using a grid factor of 0.569 kg/kWh (SONABEL 2018 mix, Nana et al.) and a PV life-cycle factor of 0.048 kg/kWh (IPCC AR5, 2014).
- Battery: 42.5 kWh = 34 / 0.8 depth of discharge, 1 day of autonomy. This is a design assumption, not an external source.

Questions for the brainstorming session

- Is the canvas built per module or for the whole project?
- Who is the final customer of the chain: the farm operator or the German buyer?
- Is energy an internal cost or a service billed to other modules?
- Which module carries the generator fuel cost: Energy Supply or Machinery?